Monthly Intelligence Report

Edition 003 — The Tejo Valley Corridor

Where the energy transition is outrunning the grid. edition · June 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

10,191

168 HV · 708 MV · 9,315 MV/LV

GRID LINES MAPPED

~11,043

~42,000 km coverage

MC SIMULATIONS

78.9 M

10,000 per substation

PUBLIC DATA SOURCES

28+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 78.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 10,191 substations and ~11,043 grid lines spanning ~105,000 km.

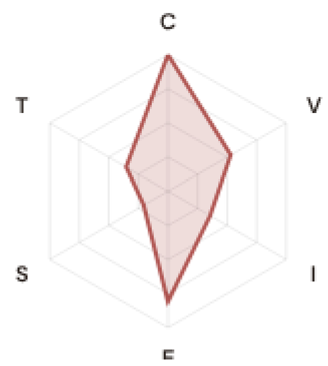
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Portugal's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Posto Açores 234

Açores, Açores · 60 kV

0.934	Critical	0.081	99th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 1.000 / 0.30 (333%)
- **I** Infrastructure: 0.355 / 0.25 (142%)
- **S** Saturation: 0.204 / 0.20 (102%)
- **V** Voltage: 0.530 / 0.10 (530%)
- **E** Economic: 0.806 / 0.10 (806%)
- **T** Transition: 0.352 / 0.05 (705%)

MODIFIERS

- **R3 Consequence:** 0.937 ↓ dampens
- **R4 Graph Criticality:** 0.950 ↓ dampens
- **R6a Restoration:** 1.100 ↑ amplifies
- **R6b Seismic:** 1.583 ↑ amplifies
- **R7 Digital Readiness:** 1.014 ↑ amplifies

This month's spotlight — automatically selected for June 2026 — is **Posto Açores 234** in Açores, Açores. With an R_final of 0.934 (Critical band, 99th fleet percentile), its risk profile is dominated by the E (Economic) component at 806% of its weight ceiling, followed by T (Transition) at 705%. Modifiers collectively shift R_base by 60.9%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Portugal Numérique / Portugal 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

1481

High/Critical + R7 < median

AVG R7 MODIFIER

0.9999

Range [0.99, 1.05]

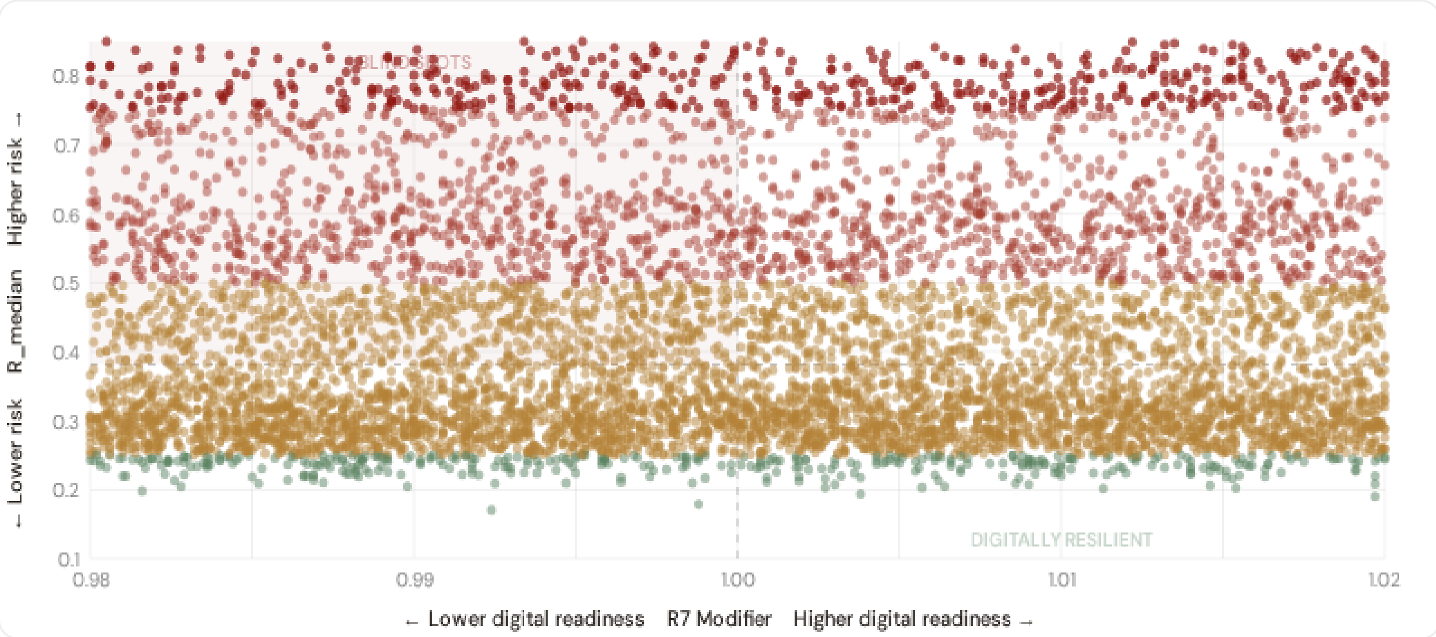
HIGH-CONSEQUENCE SUBS

2,553

25.1% · R3 ≥ 1.07

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **1481 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0000). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Centro (949), Alentejo (204), Norte (170)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh (EUR)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

2,553 substations (25.1%) High/Critical: **0** (0.0%) Avg R: **0.289** Avg E: **0.445** / 0.10

Low 318 Med 2235 High 0 Crit 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh (EUR)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

2,504 substations (24.6%) High/Critical: **800** (31.9%) Avg R: **0.416** Avg E: **0.666** / 0.10

Low 176 Med 1528 High 635 Crit 165

🏪 **Commercial / SME-Dense**

Est. VoLL: €3–8/kWh (EUR)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

2,538 substations (24.9%) High/Critical: **1310** (51.6%) Avg R: **0.527** Avg E: **0.747** / 0.10

Low 0 Med 1228 High 962 Crit 348

🌾 **Light / Agricultural / Rural**

Est. VoLL: €1–3/kWh (EUR)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

2,596 substations (25.5%) High/Critical: **979** (37.7%) Avg R: **0.525** Avg E: **0.710** / 0.10

Low 0 Med 1617 High 569 Crit 410

Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **0 substations** combine capital-intensive economic fabric (R3 ≥ 1.07) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Norte (1635), Área Metropolitana de Lisboa (918)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 15.6%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Distrito Digital Readiness Ranking

Distritos ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify distritos combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Beja ⚠ high R	0.9990
2	Évora ⚠ high R	0.9991
3	Aveiro	0.9993
4	Castelo Branco ⚠ high R	0.9993
5	Leiria ⚠ high R	0.9994
6	Viseu ⚠ high R	0.9994
7	Vila Real ⚠ high R	0.9997
8	Coimbra ⚠ high R	0.9997
9	Faro ⚠ high R	0.9998
10	Braga	0.9998
11	Lisboa	0.9998
12	Guarda ⚠ high R	0.9998
13	Porto	1.0000
14	Setúbal ⚠ high R	1.0001
15	Madeira ⚠ high R	1.0002

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Beja** has the lowest average R7 (0.9990), while **Açores** leads at 1.0024 — a spread of 0.0034 across the R7 range. **9 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (June 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9999, median = 1.0000; 0 substations classified HIGH cyber-exposure; 1481 blind-spot substations (High/Critical risk + below-median R7); 417 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **28 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

28

95 variables

WEEKLY / MONTHLY

4

High-frequency feeds

QUARTERLY / ANNUAL

17

Regular refresh cycle

STATIC / DERIVED

7

Standards + scenarios

Data Source Registry — June 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
OMIE	OMIE (Operador del Mercado Ibérico de Energía)	MONTHLY	MIBEL zone	2 vars
DGEG	DGEG (Direção-Geral de Energia e Geologia)	QUARTERLY	Concelho	5 vars
MOBI.E	MOBI.E (Rede de Mobilidade Elétrica)	QUARTERLY	Concelho	1 var
ERSE	ERSE (Entidade Reguladora dos Serviços Energéticos)	ANNUAL	Distrito	8 vars
INE	INE (Instituto Nacional de Estatística)	ANNUAL	Distrito	8 vars
ADENE	ADENE (Agência para a Energia)	ANNUAL	Distrito	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
IPMA	IPMA (Instituto Português do Mar e da Atmosfera)	ANNUAL	Distrito	2 vars
APA	APA (Agência Portuguesa do Ambiente)	ANNUAL	Distrito	1 var
AT	AT (Autoridade Tributária e Aduaneira)	ANNUAL	Distrito	1 var
ICNF	ICNF (Instituto da Conservação da Natureza)	ANNUAL	Distrito	1 var
DGEEC	DGEEC (Direção-Geral de Estatísticas)	ANNUAL	Distrito	2 vars
E-REDES	E-REDES Open Data	ANNUAL	Distrito	2 vars
CCDR	CCDR (Comissões de Coordenação Regional)	ANNUAL	Distrito	2 vars
LNEG	LNEG (Laboratório Nacional de Energia e Geologia)	STATIC	Distrito	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars

GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
ISO-9223	ISO 9223 Corrosion	DERIVED	Distrito	1 var
REN	REN (Redes Energéticas Nacionais)	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
IPMA-W	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (2 sources) — Grid load + procurement data

DGEG Registry · OMIE MIBEL

Quarterly (2 sources) — DER registry + business + macro

MOBI.E · DGEG Licensing

Annual (15 sources) — Core grid quality + demographics

ERSE · INE · DGEEC · E-REDES · ADENE · AT · EEA · Eurostat · DESI · CCDR · IPMA · APA · ICNF · JRC · REN Annual

Static (7 sources) — Standards + scenarios + seismic

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · LNEG · ISO 9223 · ENTSO-E

C.2 Fleet Score Snapshot

FLEET SIZE
10,191
substations scored

MEDIAN R
0.381
P5=0.251 · P95=0.800

HIGH + CRITICAL
3089
30.3% of fleet

HIGH CONFIDENCE
100%
10191 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 494 (4.8%) · Medium 6608 (64.8%) · High 2166 (21.3%) · Critical 923 (9.1%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **10,191 substation scores remain unchanged** this edition. The fleet median R of 0.381 (mean 0.440) reflects a distribution skewed toward the lower bands, with 4.8% in Low and 64.8% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Distrito-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — AT + CCDR + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — INE net migration	§5
L5	DATA	95/95 variables operational (100%). 28 data sources total.	§8, §12
G1	DATA	d01 REN upgraded to real-time API — grid status data	§12
G2	DATA	d05 OSM upgraded to Overpass API — 10,191 real substations	§12
G3	DATA	d06 IPMA seismic monitoring — real-time PGA + historical catalog	§12
G4	NEW	R6b Seismic modifier — variable α by distrito (Açores 0.70, Algarve 0.55, Lisboa 0.45)	§5
G5	NEW	Network & Topology layer (H): 7 variables — REN grid plan, ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Tejo Valley Corridor: Where Seismic Risk Meets Energy Transition Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Tejo Valley corridor · **002** DER saturation hotspots (Alentejo solar belt) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan-Rural economic impact disparity · **005** Seismic risk and Açores isolation · **006** Infrastructure age vs smart metering rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Distrito Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Tejo Valley, Why Now

TEJO VALLEY SUBSTATIONS

88

0 EHV · 88 HV

HIGH BAND

0 + 88

100.0% of corridor

CORRIDOR MEDIAN R

0.946

Fleet median: 0.381

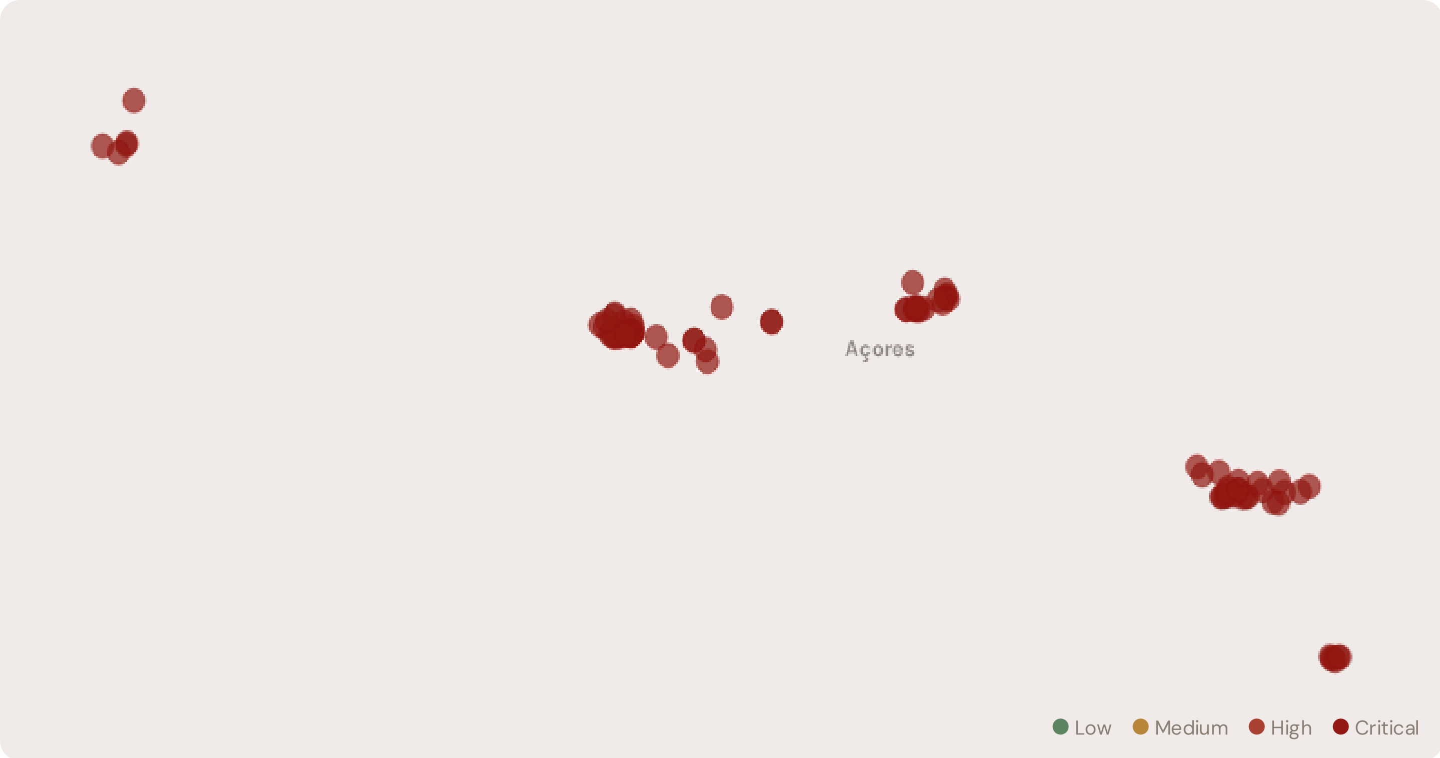
WORST DISTRITO

Açores

Avg R: 0.947 · 88 substations

The Açores hosts **88 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **0 substations (0.0%)** are classified High and **88** Critical, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.381. The corridor median R of 0.946 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Posto Açores 217	Açores	1.000	Critical	1.000	0.674	0.271	0.909	0.349	0.570	
Subestação do Aeroporto	Açores	1.000	Critical	1.000	0.543	0.291	0.797	0.357	0.328	
Subestação da Central Hídrica de Além Fazenda	Açores	1.000	Critical	1.000	0.576	0.314	0.747	0.295	0.469	
Posto Açores 1591	Açores	1.000	Critical	1.000	0.731	0.337	0.685	0.488	0.517	
Posto Açores 1592	Açores	1.000	Critical	1.000	0.621	0.348	0.859	0.249	0.477	
Posto Açores 1614	Açores	1.000	Critical	1.000	0.678	0.311	0.733	0.395	0.512	
Subestação do Parque Eólico dos Graminhais	Açores	1.000	Critical	1.000	0.587	0.290	0.739	0.412	0.543	
Subestação das Quatro Ribeiras	Açores	1.000	Critical	1.000	0.464	0.305	0.770	0.231	0.313	
Subestação da Central	Açores	1.000	Critical	1.000	0.560	0.297	0.703	0.457	0.398	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Fotovoltaica do Aeroporto										
Subestação de São Roque	Açores	1.000	Critical	1.000	0.688	0.343	0.765	0.330	0.461	

… 78 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — AÇORES VS FLEET

- C — Continuity: 1.000 vs 0.415 (+141%)
- I — Infrastructure: 0.307 vs 0.276 (+11%)
- S — Saturation: 0.310 vs 0.392 (-21%)
- V — Voltage: 0.523 vs 0.543 (-4%)
- E — Economic: 0.795 vs 0.642 (+24%)
- T — Transition: 0.434 vs 0.490 (-11%)

MODIFIER AVERAGES — AÇORES VS FLEET

R3 Consequence	0.960	fleet 1.040	↓
R4 Graph Crit.	0.968	fleet 0.955	↓
R6a Restoration	1.100	fleet 0.960	↑
R6b Seismic	1.583	fleet 1.099	↑
R7 Digital Read.	1.002	fleet 1.000	→

The dominant risk driver across the Açores corridor is **T (Transition)**, averaging 0.434 against a fleet average of 0.490 — occupying 867% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.795 vs fleet 0.642 (795% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.960 (fleet: 1.040), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.583, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Distrito Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Açores 88 subs · avg R = 0.947 · H:0 C:88 M:0

The region regions-level breakdown reveals significant intra-regional variation. **Açores** leads with a mean R of 0.947 across 88 substations, while **Açores** scores 0.947 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Açores is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

88 Açores substations score R ≥ 0.650, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Açores sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

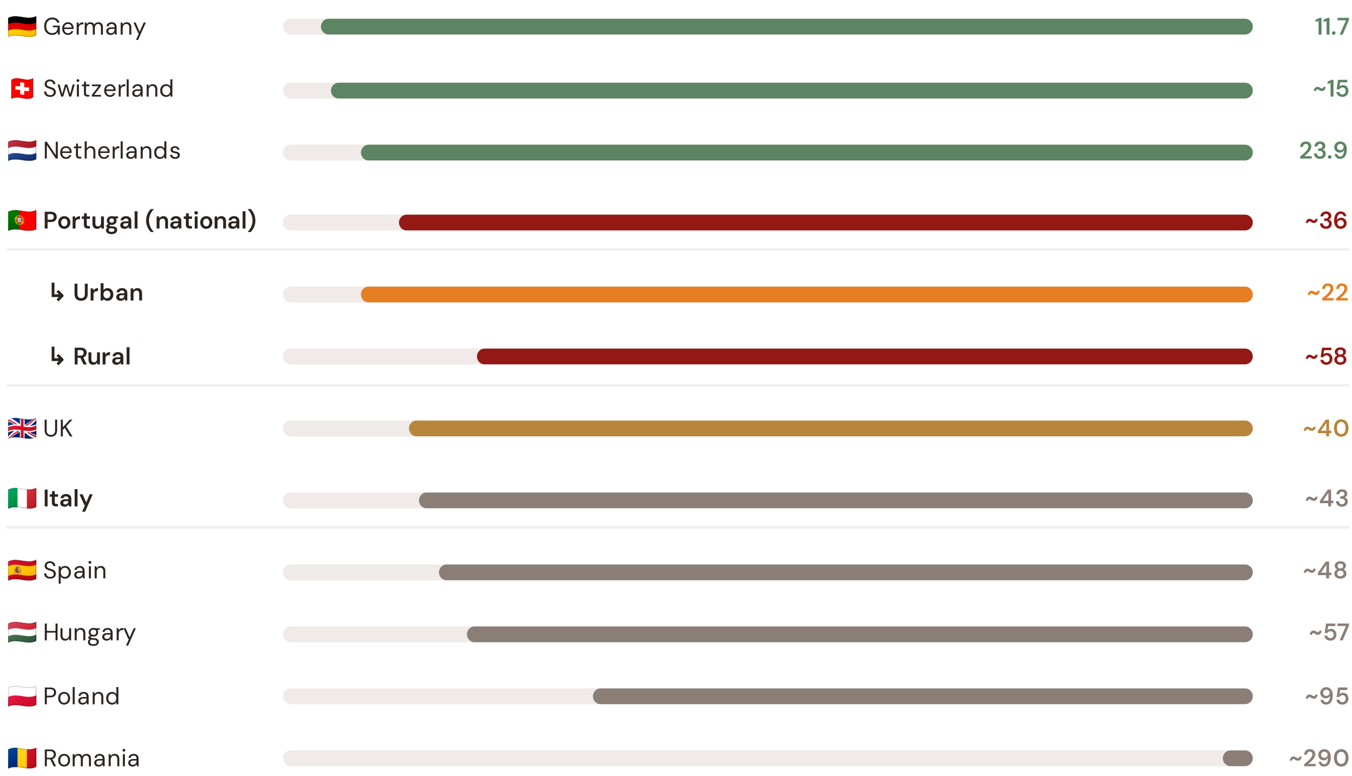
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Tejo Valley corridor analysis hinges on Portugal's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · June 2026

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER 7th Benchmarking Report (2022); ERSE TIQE (2023); Bundesnetzagentur (2024). Portugal sub-categories from ERSE territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The Açores corridor deep-dive shows **88 substations** (of 88) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 1.000 is **2.4× the fleet average** of 0.415. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 7 Portuguese Regiões

Where does the Tejo Valley sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

The Açores ranks **#1 of 7 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Açores, Alentejo, Algarve, while the three most resilient are Área Metropolitana de Lisboa, Norte, Madeira. 5 of 7 score above the fleet average of 0.440. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the E-REDES/REN and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DSO/TSO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 10,191 substations with complete feature vectors. Average CI_width of 0.046 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 494 Low, 6608 Medium, 2166 High, 923 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

MOBI.E, DGEG Licensing are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **ERSE (Entidade Reguladora dos Serviços Energéticos)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

Viseu: Highest Risk Concentration

Viseu has the highest concentration of High/Critical substations: **130 of 130 (100%)** are in the upper risk bands — avg R = 0.675. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Portuguese utility, ERSE, REN, E-REDES, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 003

Published 11 June 2026 · SSI Index v4.0.2 · 10,191 substations · ~11,043 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5